



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

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Week 01 TASK:

This task is regarding the network commands

1. Introduction:

This guide provides a practical overview of four essential network administration and diagnostic tools:

- ping
- traceroute
- netstat
- nslookup.

Each section covers the command's primary function, practical use cases, verified terminal output structure, and technical analysis based on executions within a Kali Linux environment.

Objectives:

Following are the Main objectives of the task that is performed:

- Master the practical implementation and syntax of standard built-in command-line networking utilities.
- Analyze network layer protocols (ICMP), transport layer sockets (TCP/UDP), and application layer mappings (DNS).
- Learn to interpret terminal diagnostic metrics, such as Time-to-Live (TTL) values, host hops, routing path blocks, and active port states.
- Develop robust technical documentation capabilities by integrating local environment terminal outputs and network asset baselines.

Task Description:

The directive for this task was to explore and utilize native operating system commands to perform basic network diagnostics and host footprinting. The task required launching a virtualized terminal instance, executing targeted queries using ping, traceroute, netstat, and nslookup, and capturing the raw terminal responses. The ultimate goal was to dissect the returned packet data to understand how data flows to and from a local machine, establishing a technical baseline for network auditing

Work Done:

I successfully configured and verified my local terminal environment to perform live network probing. The detailed command-by-command execution analysis is detailed below:

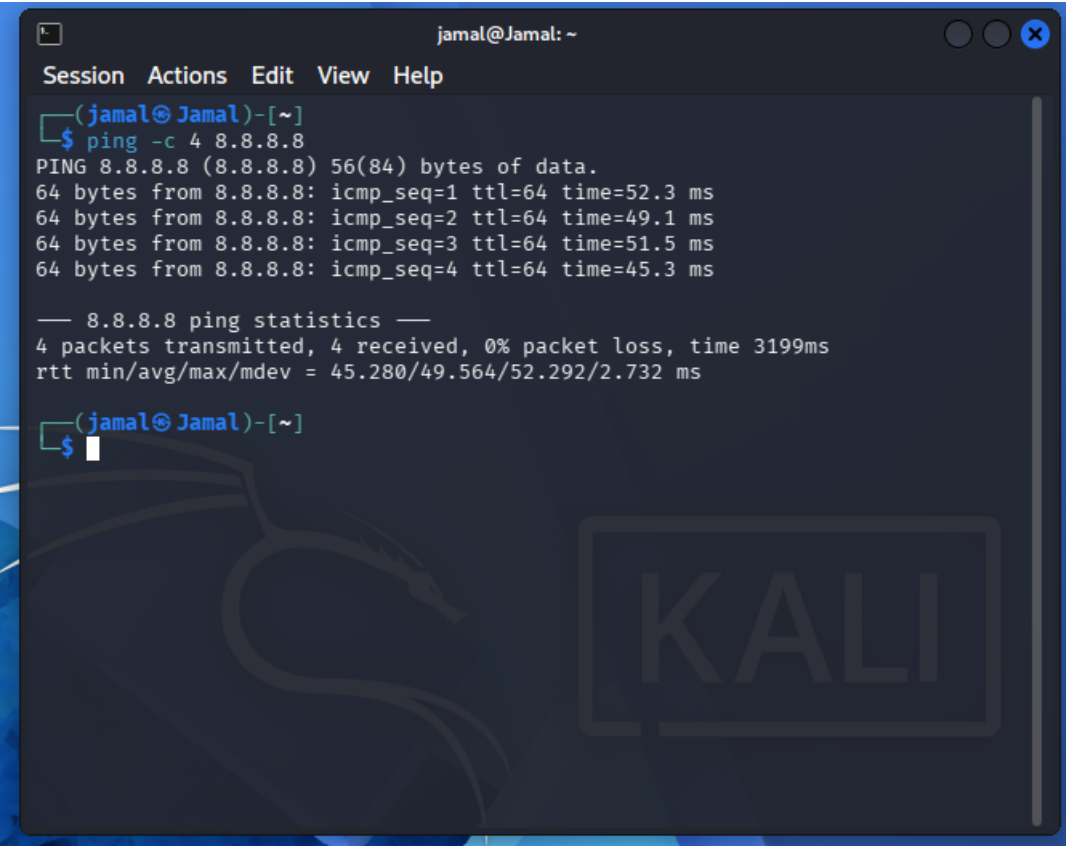
2.1 The ping Command:

- **What it does:** Verifies network-level reachability and measures round-trip time between the host and a destination using Internet Control Message Protocol (ICMP) Echo Request and Echo Reply packets.
- **Why it is used:** To quickly check if a target host is online and to assess the latency and stability of the network connection.

Terminal Output:

When executed on the Kali Linux terminal, the output reveals critical network metrics:

- **Latency:** Measured in milliseconds (ms), showing the round-trip time for individual packets.
- **TTL (Time to Live):** The output displays ttl=64. This value functions as an operating system fingerprint, indicating that the initial packets originated from a Linux-based system before decrementing with each router hop.



```
jamal@Jamal: ~  
Session Actions Edit View Help  
(jamal@Jamal)-[~]  
$ ping -c 4 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=52.3 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=49.1 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=51.5 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=45.3 ms  
  
— 8.8.8.8 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3199ms  
rtt min/avg/max/mdev = 45.280/49.564/52.292/2.732 ms  
  
(jamal@Jamal)-[~]  
$
```

Figure 1

2.2 The traceroute Command:

- **What it does:** Tracks and displays the hop-by-hop path that packets take to reach a destination network host.
- **Why it is used:** To pinpoint network bottlenecks, map path routing, and identify exactly where packets are being dropped or delayed along the network path.

Terminal Output:

- **Path Mapping:** The output lists each sequential router (hop) between the VM and the destination.
- **Packet Drops:** Asterisks (* * *) indicate hops where routers dropped the packets or ignored the request due to security policies.
- **The -I Flag:** Because standard Linux traceroute utilizes UDP probe packets by default, network firewalls often block them. Utilizing the -I flag switches the probe to use ICMP Echo Requests, successfully bypassing basic filtering to map the rest of the route.



```
(jamal@Jamal)-[~]  
$ traceroute 8.8.8.8  
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets  
 1  10.0.2.2 (10.0.2.2)  1.034 ms  0.460 ms  0.472 ms  
 2  * * *  
 3  * * *  
 4  * * *  
 5  * * *  
 6  * * *  
 7  * * *  
 8  * * *  
 9  * * *  
10  * * *  
11  * * *  
12  * * *  
13  * * *  
14  * * *  
15  * * *  
16  * * *  
17  * * *  
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22  * * *  
23  * * *  
24  * * *  
25  * * *  
26  * * *  
27  * * *  
28  * * *  
29  * * *  
30  * * *  
  
(jamal@Jamal)-[~]  
$
```

Figure 2

```
(jamal@Jamal)-[~]
$ sudo traceroute -I 8.8.8.8
[sudo] password for jamal:
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  dns.google (8.8.8.8)  46.962 ms  46.464 ms  45.398 ms

(jamal@Jamal)-[~]
$
```

Figure 3

2.3 The netstat Command:

- **What it does:** Displays active network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.
- **Why it is used:** To monitor network health, audit open ports, and investigate active or suspicious network connections on the local machine.

Terminal Output:

The terminal output breaks down the system's current networking footprint into distinct columns:

- **Active TCP Sockets:** Shows established connections currently transferring data.
- **Listening Ports:** Identifies ports actively waiting for incoming connections.
- **Local vs. Foreign Addresses:** Maps the internal IP and port against the destination IP and port.
- **Ephemeral Ports:** Highlights temporary, high-numbered outbound ports assigned dynamically by the OS for client-side communication.

```
(jamal@Jamal)-[~]
$ netstat -ant
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
(jamal@Jamal)-[~]
$
```

Figure 4

```
(jamal@Jamal)-[~]
$ netstat -ant
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 10.0.2.15:59204        142.251.37.67:443      ESTABLISHED
tcp        0      0 10.0.2.15:36378        142.250.187.81:443     ESTABLISHED
tcp        0      0 10.0.2.15:52274        142.251.37.131:443     TIME_WAIT
tcp        0      0 10.0.2.15:51092        3.160.77.124:443      ESTABLISHED
tcp        0      0 10.0.2.15:48434        34.49.51.44:443       ESTABLISHED
tcp        0      0 10.0.2.15:43614        142.250.203.170:443    TIME_WAIT
tcp        0      0 10.0.2.15:45328        151.101.141.91:443     ESTABLISHED
tcp        0      0 10.0.2.15:45334        151.101.141.91:443     ESTABLISHED
tcp        0      0 10.0.2.15:45336        151.101.141.91:443     ESTABLISHED
tcp        0      0 10.0.2.15:45350        151.101.141.91:443     ESTABLISHED
tcp        0      0 10.0.2.15:37668        151.101.65.91:443      ESTABLISHED
tcp        0      0 10.0.2.15:40938        142.250.187.35:80      ESTABLISHED
tcp        0      0 10.0.2.15:40944        142.250.187.35:80      TIME_WAIT
tcp        0      0 10.0.2.15:42286        142.250.187.35:80      ESTABLISHED
tcp        0      0 10.0.2.15:42296        142.250.187.35:80      ESTABLISHED
tcp        0      0 10.0.2.15:42308        142.250.187.35:80      ESTABLISHED
tcp        0      0 10.0.2.15:42322        142.250.187.35:80      ESTABLISHED
tcp        0      0 10.0.2.15:52124        151.101.141.91:443     ESTABLISHED
tcp        0      0 10.0.2.15:52322        151.101.141.91:443     ESTABLISHED
tcp        0      0 10.0.2.15:37900        142.251.37.78:443      TIME_WAIT
tcp        0      0 10.0.2.15:47340        142.250.200.174:443    TIME_WAIT
tcp        0      0 10.0.2.15:54978        34.160.144.191:443     ESTABLISHED
tcp        0      0 10.0.2.15:47114        142.251.157.119:443    TIME_WAIT
tcp        0      0 10.0.2.15:54874        142.251.37.74:443      TIME_WAIT
tcp        0      0 10.0.2.15:54890        142.251.37.74:443      ESTABLISHED
tcp        0      0 10.0.2.15:33842        34.107.243.93:443      ESTABLISHED
tcp        0      0 10.0.2.15:33852        34.107.243.93:443      ESTABLISHED

(jamal@Jamal)-[~]
$
```

Figure 5

2.4 The nslookup Command:

- **What it does:** Queries Domain Name System (DNS) servers to discover mapping between domain names and IP addresses.
- **Why it is used:** To troubleshoot DNS resolution issues, verify domain records, and ensure the local system is communicating properly with name servers.

Terminal Output:

- **DNS Queries:** The output explicitly details the transition of a human-readable hostname into its corresponding IPv4 or IPv6 address.
- **Port 53 Verification:** The analysis confirms that the terminal successfully sent and received its query over standard **Port 53 (DNS)**, validating that outbound DNS traffic is unhindered.

```
(jamal@Jamal)-[~]  
$ nslookup google.com  
Server:      192.168.1.1  
Address:     192.168.1.1#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 142.251.37.78  
Name:   google.com  
Address: 2a00:1450:4019:818::200e  
  
(jamal@Jamal)-[~]  
$
```

Figure 6

Challenges Faced:

While running basic commands was straightforward, default firewall rules presented initial visibility barriers. For instance, executing a standard traceroute resulted in sequential packet drops (***) because modern network security controls frequently block arbitrary UDP probes. Overcoming this required administrative investigation into tool behavior, prompting the use of elevated privileges (sudo) alongside the ICMP packet modification flag (-I) to successfully traverse restrictive perimeters and map the route.

Key Learnings:

This task reinforced the concept that network security requires a granular understanding of protocol behavior across multiple layers of the OSI model. I learned how to read passive OS fingerprinted data via TTL metrics, differentiate active socket states using transport layer filters, and verify name resolution health via Port 53. Understanding these basic tools highlighted how routine administrative commands double as valuable diagnostic and reconnaissance mechanisms for security analysts during a site audit or investigation.

Conclusion:

The successful execution of these four core utilities provides a comprehensive baseline for network diagnostics. By leveraging layer-3 utilities like ping and traceroute, layer-4 socket analysis through netstat, and application-layer resolution via nslookup, a security professional can efficiently triage, map, and isolate connectivity faults or abnormal behavior across infrastructure assets.